



Cost consideration of cloud services

Hello, future cloud pro!

Thanks for being with me for this rundown on the key features affecting cloud costs: resource provisioning, storage, and running queries.

Let's start with an example.

Say you're headed to the market, so you create a shopping list.

You think about what you'll need in the coming week, then write down exactly those items.

The list helps ensure you don't overspend on items you don't need, or that might go bad later.

Well, managing the costs of cloud data analytics is kind of like that.

The key is to be a super savvy shopper, knowing exactly what resources you need, and how much.

This saves money and prevents waste.

The first method cloud professionals use to achieve these goals is **resource provisioning**.

This is the process of a user selecting appropriate software and hardware resources and the cloud service provider setting them up and managing them while in use.

The resource provisioning process occurs through one of three delivery models: **advanced provisioning**, **dynamic provisioning**, and **self-provisioning**.

Each delivery model is different based on the types of resources an organization buys, how and when it receives these resources, and how it pays for them.

In **advance provisioning**, the user signs a formal agreement with the cloud service provider and either pays a set price or is billed monthly.

Then, the provider gathers the agreed upon resources and delivers them to the user.

In **dynamic provisioning**, resources are adjusted based on the user's changing needs and they're only charged for what they use.

This means they can easily scale up or down based on usage demands.

With **self-provisioning**, also known as cloud self-service, the user purchases resources from the cloud provider through a website or online portal, and then the resources are quickly made available for the user, usually within hours or even minutes!

Although **users only pay for the resources they use**, how they choose to receive these resources affects how much they pay.

Payment can be arranged with one of three rates: **fixed**, **pay-as-you-go**, or **instant purchase**.

Now let's move to **storage costs**.

Storage is ranked as one of the top three cloud expenses for many organizations, and the demand for more storage capacity only continues to grow.

Storage costs vary based on **data storage**, **data processing**, and **network use**.

As the term suggests, **data storage** is the amount of data kept in storage.

In the cloud profession, we refer to these as buckets.

Exactly how much an organization pays for this storage can actually change based on where those

buckets are located in the world and the type, or "class," of the data being stored.

For example, **coldline storage** data is great for data you read or change once a quarter, but **archive storage**

data keeps data that is only meant for backup or archiving purposes, and it's the cheapest form of data storage.

Now, **data processing** is the **step where raw data is cleaned, organized, and changed into a format for easy analysis.**

Data processing can significantly affect storage costs in several ways, **like the more data you process the more storage space you'll need.**

And **faster data processing requires more advanced storage solutions.**

Finally, **network use** refers to the amount of data that is read or moved between storage buckets.

Ok, let's dive into the costs associated with running queries on stored data.

One interesting point to understand is that most cloud service providers charge for query processes

based on the amount of data processed, not the amount returned after the query is run.

So, every time a cloud data analyst runs a query to retrieve, update, or analyze data, a bill is generated based on the total amount of data that query processes.

With Google BigQuery, for example, there's a choice of two pricing models for running queries: on-demand pricing and capacity pricing.

On-demand pricing is based on the amount of data, in bytes, that each query processes.

The on-demand pricing model is ideal for users whose compute and storage needs fluctuate based on business priorities.

And capacity pricing is based on the computing power used to run queries over time, measured in virtual central processing units, called slots.

The capacity pricing model is ideal for users who want a more predictable, controllable cost for their queries.

As a cloud data analytics professional, you can bring a ton of value to your employer

by using this knowledge to get the best performance and value from cost-effective cloud resources and services.

Best of all, you'll be helping them optimize resources, processes, and the bottom line!